

Project Specifications Report

1. Introduction

1.1 Description

The Anomaly-Based Accident Detection and Faulty Vehicle Identification System is a high-level computer vision and deep learning project designed to automate the lifecycle of traffic accident analysis. The system functions by continuously monitoring real-time video streams to identify irregular traffic behaviors through a sophisticated Anomaly Detection framework. Unlike conventional surveillance systems, this project does not merely detect an incident; it extends the analysis by autonomously categorizing the collision type, pinpointing the primary faulty actor based on spatio-temporal deviation, and documenting the responsible vehicle's identity via License Plate Recognition (LPR). The system aims to provide objective, immediate, and verifiable forensic data for law enforcement and insurance providers, minimizing human error and enhancing road safety management through a fully automated diagnostic pipeline.

1.2 Constraints

Environmental Conditions: The accuracy of the detection and identification modules may be affected by extreme weather conditions (heavy fog, torrential rain, or snow) and low-light environments which can degrade image quality.

Hardware Limitations: Real-time processing of multiple high-resolution deep learning models requires significant computational power (GPU), limiting the system's performance on low-end edge devices without optimization.

Occlusion and Camera Angle: The system's ability to identify faulty vehicles and read license plates is constrained by the camera's field of view; significant occlusions by larger vehicles or suboptimal camera angles may hinder data extraction.

Data Privacy: The collection and storage of license plate data must comply with local data protection regulations (e.g., KVKK or GDPR), necessitating secure encryption and access control protocols.

Processing Latency: There is a trade-off between the complexity of the video classification models and the speed of real-time reporting, requiring a balance to ensure timely incident alerts.

1.3 Professional and Ethical Issues

Data Privacy and Confidentiality: The system must adhere to strict data protection laws (e.g., GDPR, KVKK) as it processes sensitive information such as license plates and vehicle movements. All captured data must be encrypted and accessible only by authorized personnel.

Algorithmic Bias and Fairness: The models must be trained on diverse datasets to ensure that detection and liability assessment are not biased toward specific vehicle types, colors, or environmental conditions.

Accountability and Transparency: While the system provides automated liability reports, these should be treated as "decision support" rather than absolute legal verdicts. There must be a transparent mechanism for human review to prevent wrongful accusations due to technical errors.

Public Safety and Trust: The ultimate goal is to enhance public safety. Therefore, the system must maintain high reliability to prevent false alarms that could lead to unnecessary panic or administrative burdens for law enforcement.

2. Requirements

The system must satisfy a integrated set of operational and performance requirements to ensure effective accident detection. Primarily, it shall maintain continuous real-time monitoring to track vehicle trajectories and identify spatio-temporal anomalies such as collisions or sudden swerves. Upon detection, the system is required to automatically extract event clips, classify the accident type, and perform causality-based fault attribution. Furthermore, the system must achieve high detection accuracy (Pd) while minimizing false alarms (Fa). From a technical perspective, the solution must provide low-latency processing, ensure data integrity through encryption, and remain scalable for multi-camera deployments. All operations must comply with data privacy regulations while maintaining 24/7 reliability under varying environmental conditions.

3. References

ACM Code of Ethics and Professional Conduct

The Software Engineering Code of Ethics, IEEE Computer Society

IEEE Code of Ethics

Computer and Information Ethics, Stanford Encyclopedia of Philosophy

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